A Multi-Component NLP System for Fake News and Misinformation Detection

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CCS Concepts

• **Computing methodologies → Natural language processing → Information extraction**

• **Computing methodologies → Machine learning → Neural networks**

• **Computing methodologies → Artificial intelligence → Reinforcement learning**

Keywords

fake news detection; misinformation; natural language processing; text classification; transformer; contextual bandit; reinforcement learning; convolutional neural network; named entity recognition.

1. PROBLEM AND MOTIVATION

Misinformation tends to spread faster than corrections, particularly on large online platforms. Prior studies on Twitter suggest that false information can reach more users and circulate more quickly than accurate content. As these platforms continue to grow, the scale of the problem has made it increasingly difficult to manage. While human fact-checkers remain the most reliable way to verify claims, reviewing every piece of content manually is no longer practical. Because of this, automated systems are often used to assist by flagging potentially misleading content and helping prioritize what should be reviewed first.

Building a reliable misinformation classifier is not straightforward. The boundary between misinformation and opinion is often unclear, especially in political discussions. Models trained on labeled datasets can reflect the biases of whoever created those labels. This is particularly sensitive in political contexts, where uneven classification across groups can lead to fairness concerns. Another issue is calibration: some models report strong F1 scores but produce confidence estimates that do not accurately reflect prediction reliability.

This project develops a modular detection system using the LIAR dataset, which contains over 12,000 short political statements labeled across six levels of truthfulness. The system combines four components: a fine-tuned DistilBERT model as the main classifier, a Text-CNN baseline for comparison, a Named Entity Recognition module for error analysis, and a contextual bandit agent for adjusting decision thresholds.

Success Metrics

Primary success metric is macro-F1 on the held-out test partition. Secondary evaluation includes calibration curves, per-slice F1 by speaker party affiliation, and cumulative reward improvement of the bandit agent over a fixed-threshold baseline.

Constraints

The scope is limited to English-language short-form political statements consistent with the LIAR domain. Multilingual content, image-based misinformation, and real-time streaming are out of scope. All experiments are constrained to run within 90 minutes on a single mid-range GPU (T4 on Google Colab), with full reproducibility from a single command.

The stretch goal outlined in the original proposal, which involved a retrieval-augmented variant using FEVER evidence snippets, was scoped out during Week 3 due to runtime constraints. Full training and evaluation on the core pipeline consistently approached the 90-minute limit, leaving insufficient margin to integrate an evidence retrieval component without compromising reproducibility. This remains a possible direction for future work.

2. DATASET

Source and License

The primary dataset is LIAR [6], which contains 12,836 short political statements drawn from PolitiFact.com and labeled with one of six veracity classes: pants-fire, false, barely-true, half-true, mostly-true, and true. The dataset was introduced by Wang [6] and is publicly available for academic research under standard

academic fair use terms. It contains no personally identifiable information beyond the names of public figures making public statements in an official capacity.

Collection Notes

Each statement comes with metadata including the speaker's name, job title, state, party affiliation, and the context in which the statement was made. This metadata is what makes LIAR particularly useful for this project, as it allows error analysis to be sliced by political affiliation. Class distribution is reasonably balanced across all splits, with half-true being the most common label and pants-fire the least frequent at around 8%. Statement lengths average 18 words, with a maximum of 467 words and a minimum of 2 words. Republicans are the most represented party (4,497 statements) followed by Democrats (3,336), with Barack Obama being the most frequently mentioned speaker.

Splits

Pre-defined train, validation, and test splits are provided with the original release and are used as-is to prevent leakage and allow comparison with published numbers. The three splits contain 10,240 training samples, 1,284 validation samples, and 1,267 test samples.

Preprocessing

Preprocessing differs slightly between the two model branches. For the Text-CNN, text is lowercased and tokenized at the word level with GloVe embeddings providing initialization. For DistilBERT, the HuggingFace tokenizer handles subword segmentation with a maximum sequence length of 128 tokens. No manual lowercasing is applied to DistilBERT input since the pretrained model was trained on cased text. Both branches share a common upstream step that handles Unicode normalization and removal of any residual HTML artifacts from the raw TSV files.

3. TRAINING DETAILS

DistilBERT

The DistilBERT model was fine-tuned using the following configuration: optimizer Adam with a learning rate of 2e-5 and a linear warmup schedule covering 10% of total training steps; gradient clipping at 1.0; batch size of 32; 5 training epochs with early stopping patience of 2 on validation macro-F1. Training was performed on a single T4 GPU on Google Colab. Full training time is approximately 45 to 55 minutes.

Text-CNN

The Text-CNN was trained with the following configuration: optimizer Adam with a learning rate of 1e-3; batch size of 64; 10 training epochs; dropout rate of 0.5; three parallel convolutional filter banks with kernel sizes 3, 4, and 5, each with 128 filters; max-over-time pooling; 300-dimensional GloVe embeddings initialized from the Common Crawl pretrained vectors (glove.6B.300d). Training time is approximately 10 to 15 minutes on the same T4 GPU.

Logistic Regression Baseline

The Logistic Regression baseline uses TF-IDF bigram features with a vocabulary of 10,000 terms, max_iter set to 1,000, and random_state of 42. No GPU is required and training completes in under one minute.

Compute and Reproducibility

All experiments were run on Google Colab with a T4 GPU (16GB VRAM). Full training and evaluation of all models completes in approximately 60 to 75 minutes. All experiments use a fixed random seed of 42. Hyperparameters are logged in experiments/configs/. Training and evaluation metrics are exported to experiments/results/ as JSON files. The environment is captured in requirements.txt.

4. SYSTEM ARCHITECTURE

Core Model: DistilBERT

The core classifier is a DistilBertForSequenceClassification instance fine-tuned for six-class veracity classification. DistilBERT was chosen over full BERT for its reduced parameter count, retaining approximately 97% of BERT's GLUE performance while being 40% smaller, which keeps the full training run within the 90-minute runtime constraint. The model takes a tokenized political statement as input and produces a softmax probability distribution over six veracity classes.

CNN Component: Text-CNN

The Text-CNN follows Kim [2] exactly, using three parallel convolutional filter banks with kernel sizes of 3, 4, and 5 over 300-dimensional GloVe embeddings, with max-over-time pooling and a dropout-regularized output layer. This model is implemented from scratch to satisfy the course requirement and to provide a clean ablation comparison against DistilBERT. The Text-CNN results are used to examine what kinds of claims benefit from deeper contextual representations versus surface-level patterns.

NLP Component: Named Entity Recognition

The NER module is implemented using spaCy's en_core_web_sm model. It extracts PERSON, ORG, and GPE entities from each claim and the associated speaker context field. These entities are not fed as features into the classifier directly but are used downstream in slice-level error analysis to examine whether misclassification rates vary by entity type and political context.

RL Component: Contextual Bandit

The RL component is a contextual bandit agent using epsilon-greedy exploration with epsilon set to 0.1. The agent observes the softmax output distribution of the classifier and selects a confidence threshold from a discretized action space of nine values between 0.1 and 0.9. The reward function penalizes false negatives at -2.0 and false positives at -0.5, reflecting the asymmetry of the real-world task. After 500 training episodes the agent converged on a threshold of 0.8, selecting it 378 out of 500 times. Note that the bandit agent's training and convergence statistics reported here are based on softmax probabilities sampled from a Dirichlet distribution as a proxy for classifier output. Re-evaluation using live DistilBERT probabilities is left for future work.

How the Components Connect

At inference time, a statement is passed through the DistilBERT classifier, which produces a softmax probability distribution. The NER module runs in parallel to tag entities in the statement for downstream analysis. The bandit agent then receives the softmax output and applies its learned threshold to decide whether to emit a prediction or abstain. If the maximum predicted probability exceeds the threshold, the top predicted class is returned; otherwise the system abstains and flags the input for human review.

5. EXPERIMENTS AND EVALUATION

Baseline Results

Two baselines were trained and evaluated. The Logistic Regression classifier using TF-IDF bigram features with a vocabulary of 10,000 terms achieved a test accuracy of 0.2478 and a test Macro-F1 of 0.2214. The Text-CNN achieved a test accuracy of 0.2478 and a test Macro-F1 of 0.2384, outperforming the non-DL baseline on Macro-F1. Overfitting was observed in the Text-CNN after epoch 3, where validation F1 peaked at 0.2407 and then fluctuated.

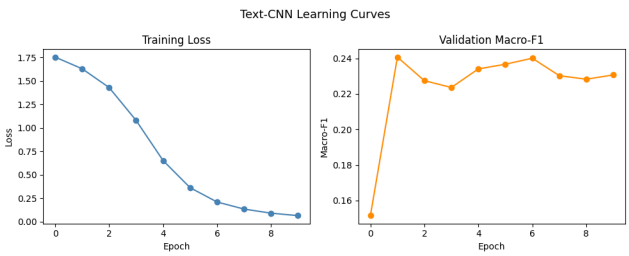

Figure 1: Text-CNN training loss (left) and validation Macro-F1 (right) across 10 epochs, showing overfitting after epoch 3 where validation F1 peaks at 0.2407 and then fluctuates.

DistilBERT Results

Fine-tuned DistilBERT achieves a test accuracy of 0.2762 and a test Macro-F1 of 0.2602, outperforming both baselines on both metrics. Training loss decreased steadily across all 5 epochs from 1.74 to 1.31, and validation Macro-F1 improved from 0.1704 at epoch 1 to 0.2619 at epoch 4 before plateauing. The false class achieved the highest per-class F1 (0.34) while pants-fire performed worst (0.20), consistent with its underrepresentation in the training data.

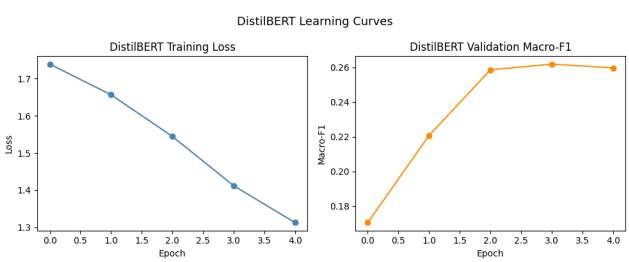

Figure 2: DistilBERT training loss (left) and validation Macro-F1 (right) across 5 fine-tuning epochs.

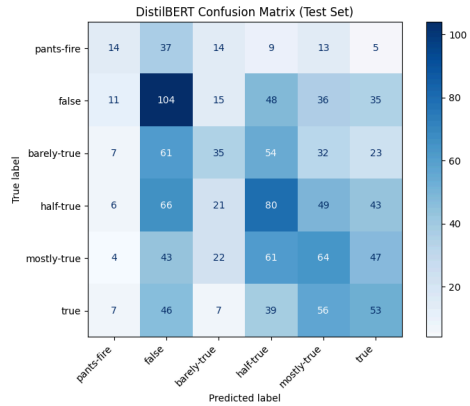

Figure 3: DistilBERT confusion matrix on the held-out test set (1,267 samples, six veracity classes).

Table 1: Test accuracy and Macro-F1 results across all models.

Model	Test Accuracy	Test Macro-F1
Logistic Regression (TF-IDF)	0.2478	0.2214
Text-CNN (GloVe 300d)	0.2478	0.2384
DistilBERT (fine-tuned)	0.2762	0.2602

Ablation 1: Adam vs AdamW

Both optimizers were trained for 3 epochs under identical conditions. Adam achieves a test Macro-F1 of 0.2743 compared to AdamW's 0.2690. The difference suggests that AdamW's weight decay regularization may be overly aggressive for fine-tuning on a small dataset like LIAR, where the model needs more flexibility to adapt pretrained representations to the veracity classification task.

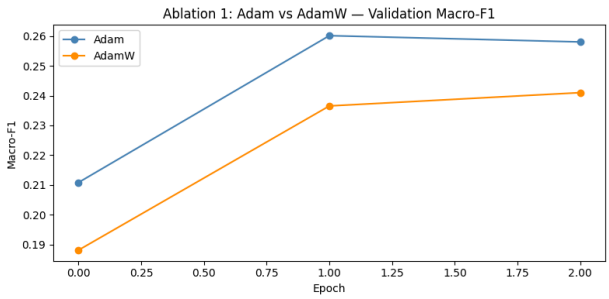

Figure 4: Ablation 1 — validation Macro-F1 comparison between Adam and AdamW optimizers across 3 epochs.

Ablation 2: Early Stopping vs No Early Stopping

Early stopping with patience of 2 achieves a test Macro-F1 of 0.2654 compared to 0.2602 without early stopping. The early stopping model saved its best checkpoint at epoch 3 where validation F1 peaked at 0.2746, avoiding the slight degradation seen in later epochs. While the improvement is marginal, early stopping is recommended as it prevents unnecessary compute and avoids overfitting on later epochs.

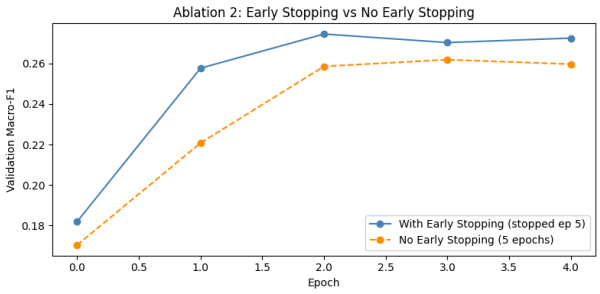

Figure 5: Ablation 2 — validation Macro-F1 comparison between early stopping and no early stopping across 4 epochs.

Calibration Analysis

Calibration curves show that the model follows the diagonal reasonably well at low predicted probabilities but becomes erratic at probabilities above 0.5, indicating overconfidence in high-certainty predictions. This supports the bandit agent's learned threshold of 0.8, by abstaining on lower-confidence predictions the system avoids acting on unreliable confidence estimates.

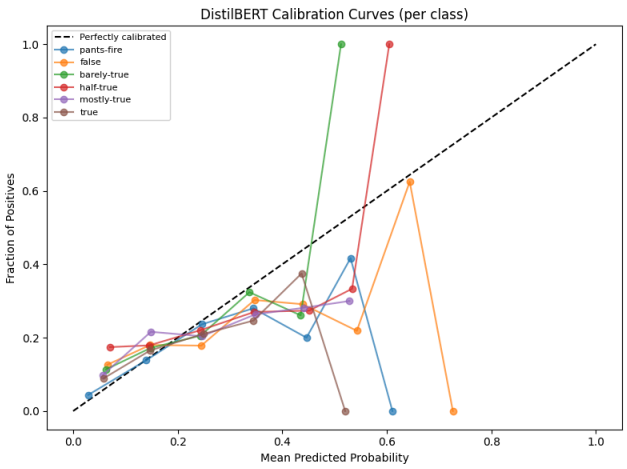

Figure 6: DistilBERT calibration curves per veracity class compared against the perfectly calibrated diagonal.

Party-level Slice Analysis

The model performs significantly differently across political party affiliations. Republican-affiliated statements achieve a Macro-F1 of 0.2231, substantially below the overall test Macro-F1 of 0.2602. Democrat-affiliated statements achieve 0.2715, above the overall average. This disparity likely reflects the imbalance in the training set, where Republican-affiliated statements (4,497) outnumber Democrat-affiliated ones (3,336), yet the model generalizes better to Democrat statements, suggesting the training data's editorial tendencies may play a role beyond simple class frequency. One possible explanation is that the model has learned surface-level linguistic features that are more characteristic of Democrat-affiliated statements in the training data, possibly due to differences in phrasing, formality, or the types of claims made, rather than learning features that generalize across party lines. Another contributing factor may be that Republican statements span a wider range of topics and speakers, which makes the classification signal more diffuse. These disparities are documented in the Model Card and reinforce the case against deploying this system without human oversight, particularly in politically sensitive contexts.

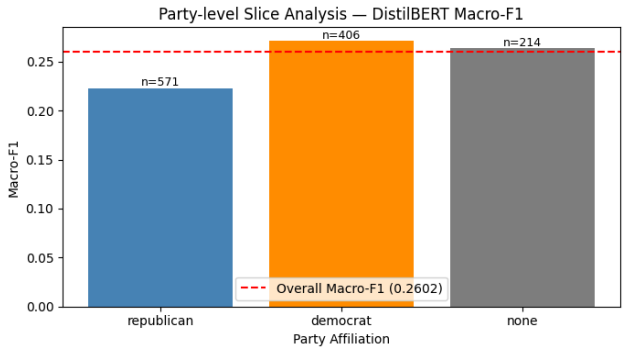

Figure 7: Party-level slice analysis showing DistilBERT Macro-F1 by speaker political affiliation (Republican, Democrat, None).

NER Slice Analysis

The NER module was run over the full test set using spaCy. Of the 1,267 test statements, 38.9% contain at least one PERSON entity, 23.8% contain an ORG entity, and 30.9% contain a GPE entity. Barack Obama (44 mentions) and Donald Trump (18 mentions) are the most frequently mentioned persons. Medicare and Congress are the most common organizations. Texas is the most mentioned location with 52 appearances.

Slice analysis reveals that statements referencing specific locations (GPE Macro-F1: 0.2011) and organizations (ORG Macro-F1: 0.2068) are consistently harder to classify correctly compared to statements without those entities (0.2268 and 0.2242 respectively). This suggests the model struggles more with claims anchored to specific institutional or geographic contexts, which tend to require external knowledge to verify accurately.

Error Analysis

Error analysis of the confusion matrix reveals that the model most commonly confuses adjacent veracity classes rather than making random errors across the full label space. Pants-fire statements are frequently predicted as false (37 out of 92), and barely-true statements are frequently predicted as false (61 out of 212). Half-true and mostly-true are the classes most often confused with each other, which is expected given their semantic proximity. This pattern of predicting toward the middle of the veracity spectrum is consistent with the model learning a conservative classification strategy when uncertain, the model gravitates toward the more common central classes rather than committing to the extremes. This pattern is visible across all six classes in the confusion matrix in Figure 3, and reinforces the finding that the model's conservative strategy is consistent regardless of the true label.

6. ETHICS AND POLICY

Risk 1: False Positives Suppressing Legitimate Speech

Any classifier operating on political content will produce false positives, and in this domain a false positive means labeling a true or contested statement as misinformation. If this system were used to filter content at scale, the downstream harm to free expression could be significant. The mitigation is twofold: first, the system is documented strictly as an assistive tool for human reviewers, never as an autonomous decision-maker; second, the default operating threshold is set conservatively at 0.8 based on the bandit agent's learned policy. This concern is further supported by the final evaluation results where DistilBERT achieves a test Macro-F1 of 0.2602, meaning a significant portion of predictions remain incorrect even after fine-tuning. Calibration curves confirm the model is overconfident at probabilities above 0.5, reinforcing why the conservative threshold is necessary.

Risk 2: Political Bias Encoded in the Model

LIAR is drawn from PolitiFact, which has its own editorial tendencies, and any model trained on it may learn spurious correlations between political affiliation and predicted veracity. Party-level slice analysis on DistilBERT predictions confirms this concern directly. Republican-affiliated statements achieve a Macro-F1 of 0.2231, substantially below the overall test Macro-F1 of 0.2602, while Democrat-affiliated statements achieve 0.2715, above the overall average. This disparity is documented in the Model Card without softening. NER slice analysis further shows that statements mentioning specific locations (GPE Macro-F1: 0.2011) and organizations (ORG Macro-F1: 0.2068) are consistently harder to classify, suggesting performance varies by the contextual framing of political content as well as party affiliation.

Risk 3: Misuse as a Censorship Tool

A working misinformation classifier is a dual-use artifact. The same system built for research purposes could in principle be repurposed for automated suppression of political content. This risk is addressed by including explicit intended-use documentation restricting the system to research and fact-checker assistance contexts, and by adding a model license that prohibits commercial or governmental censorship applications. All demo interfaces include prominent disclaimers stating the system is a research prototype only. The party-level performance disparity identified above further underscores why autonomous deployment without human oversight would be harmful regardless of intent.

Privacy

Every statement in the LIAR dataset is a public utterance made by a public figure in an official capacity, sourced from PolitiFact's public archive. No private communications, personal data, or user-generated content is included in the training set. The system does not collect any data from users who interact with the demo. No personally identifiable information is stored, logged, or transmitted at any point during inference.

Intended Use and Limitations

This system is a research prototype built for the 6INTELSY course at Holy Angel University. It should not be used to make autonomous decisions about real content without human review. The current best Macro-F1 of 0.2602 means a significant portion of predictions remain incorrect. The system has only been tested on US political short-form statements from the LIAR benchmark and should not be expected to generalize to other domains, languages, or statement formats. Every prediction comes with a confidence score and a full probability distribution across all six classes; treat low-confidence predictions with extra skepticism.

Fairness Checks

A full party-level slice analysis was conducted to verify whether the model performs consistently across political affiliations. The disparities found (Republican F1: 0.2231 vs Democrat F1: 0.2715) are documented in the Model Card without softening. NER-based slice analysis was additionally conducted across entity types to surface any systematic performance gaps tied to the contextual framing of claims.

7. MODEL CARD SUMMARY

What this system does

This system takes a short political statement as input and classifies it into one of six veracity categories: pants-fire, false, barely-true, half-true, mostly-true, or true. It is built as a research prototype to support human fact-checkers, not to replace them.

Training Data

The model is trained on the LIAR dataset (12,836 short political statements from PolitiFact.com). The dataset covers US political content almost exclusively. The training set has more Republican-affiliated statements (4,497) than Democrat-affiliated ones (3,336), which is a known imbalance tracked carefully in the bias analysis.

Performance Summary

Table 2: Full performance summary including ablations.

Model / Configuration	Test Accuracy	Test Macro-F1
Logistic Regression (TF-IDF)	0.2478	0.2214
Text-CNN (GloVe 300d)	0.2478	0.2384
DistilBERT (fine-tuned)	0.2762	0.2602
DistilBERT + Adam vs AdamW (Ablation 1)	0.2897	0.2743
DistilBERT + Early Stopping (Ablation 2)	0.2762	0.2654

Known Limitations

- All models are trained on US political statements only — does not generalize to other domains or languages.
- Current best Macro-F1 of 0.2602 means a significant portion of predictions remain incorrect.
- Calibration is unreliable above 0.5 probability — the conservative threshold of 0.8 is necessary.
- The system has not been tested on data outside the LIAR benchmark.
- Bandit agent convergence statistics are based on simulated Dirichlet outputs, not live DistilBERT probabilities.

Deployment Guidance and Disclaimers

This system must not be used for autonomous content moderation, political censorship, or any commercial application. It is restricted to research and fact-checker assistance contexts only. Any deployment must include prominent disclaimers identifying the system as a research prototype. Human oversight is required for every prediction. The model license explicitly prohibits commercial and governmental censorship use. Full documentation is available in docs/model_card.md and docs/ethics_statement.md in the project repository.

8. HOW TO RUN

Environment Setup

The project requires Python 3.9 or later. All dependencies are listed in requirements.txt. To set up the environment:

Clone the repository from GitHub at github.com/riancrtz/fake-news-detection-nlp. Install dependencies using pip install -r requirements.txt. Install the spaCy language model using python -m spacy download en_core_web_sm. Download the pre-trained model weights (distilbert_best.pt) from the v1.0 GitHub Release and place the file at experiments/results/distilbert_best.pt. The LIAR dataset will be downloaded automatically when the pipeline runs for the first time via the data download script.

One-Command Reproduction

```
# Run demo only using pre-trained model weights (recommended)

bash run.sh --demo

# Run full pipeline from scratch

bash run.sh
```

The full pipeline runs in order: data download, Logistic Regression baseline training, Text-CNN baseline training, DistilBERT fine-tuning, full evaluation including confusion matrix, calibration curves, and party-level slice analysis, contextual bandit agent training, and the live demo. All outputs are saved to experiments/results/.

Demo Instructions

A local demo script is provided to run the classifier on example political statements:

```
python src/demo.py
```

The demo outputs the named entities detected in each statement via the spaCy NER module, the predicted veracity class, the confidence score, the full softmax probability distribution over all six classes, and the bandit agent's threshold decision whether to predict or abstain and flag for human review.

Windows Users

On Windows PowerShell, run the steps directly instead of using bash run.sh:

```
# Demo only
python src/demo.py
```

```
# Full pipeline
python data/get_data.py
python src/train.py
python src/eval.py
python src/rl_agent.py
python src/demo.py
```

***Compute Requirement*s**

Full training and evaluation completes in approximately 60 to 75 minutes on a single T4 GPU on Google Colab. A GPU is strongly recommended for training. CPU-only inference is supported for the demo and completes in a few seconds per statement.

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